

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Professional Training Curriculum

A structured curriculum covering Python programming, mathematics and statistics, data analysis, machine learning, deep learning, NLP, Generative AI, deployment, and a real-world AI/ML project.

PROGRAM STRUCTURE

12 Comprehensive Modules

AI / MACHINE LEARNING SYLLABUS

Module 1 — Python for AI/ML

- Python Fundamentals & Programming
- Functions, OOP & File Handling
- NumPy & Pandas
- Data Handling
- Jupyter & Google Colab

Module 2 — Mathematics & Statistics for ML

- Linear Algebra Fundamentals
- Probability & Statistics
- Descriptive & Inferential Statistics
- Data Preprocessing
- Mathematical Foundations

Module 3 — Data Analysis & Visualization

- Data Cleaning & Preprocessing
- Exploratory Data Analysis
- Matplotlib & Seaborn
- Data Visualization
- Extracting Insights from Data

Module 4 — Machine Learning Fundamentals

- AI, ML & Deep Learning Concepts
- Supervised & Unsupervised Learning
- Training, Testing & Validation
- Bias & Variance
- Overfitting & Underfitting

Module 5 — Supervised Learning

- Linear & Polynomial Regression
- Logistic Regression
- Decision Trees
- Random Forest
- KNN
- SVM

Module 6 — Unsupervised Learning

- Clustering Concepts
- K-Means
- Hierarchical Clustering
- Dimensionality Reduction
- PCA
- Anomaly Detection

Module 7 — Model Evaluation & Optimization

- Classification & Regression Metrics
- Confusion Matrix
- Precision & Recall
- Cross-Validation
- Hyperparameter Tuning
- Feature Selection
- Model Optimization

Module 8 — Deep Learning

- Neural Network Fundamentals
- ANN
- Backpropagation
- CNN for Image Processing
- TensorFlow/Keras
- Model Training

Module 9 — Natural Language Processing

- NLP Fundamentals
- Text Processing
- Tokenization
- Stop Words
- Embeddings
- Sentiment Analysis
- Text Classification
- Basic NLP Applications

Module 10 — Generative AI & Modern AI

- Generative AI Fundamentals
- Large Language Models
- Prompt Engineering
- AI APIs
- AI Application Development
- Responsible & Ethical AI

Module 11 — AI Deployment & Applications

- Model Saving & Deployment
- Flask/FastAPI
- Streamlit
- AI Dashboards
- Cloud AI Fundamentals
- Edge AI Fundamentals

Module 12 — Live AI/ML Project

- Real-World Dataset & Problem Definition
- Data Processing
- Model Development
- Model Evaluation
- Model Deployment
- Final AI/ML Project Presentation

CORE TECHNOLOGIES

Python • NumPy • Pandas • Matplotlib • Scikit-learn • TensorFlow/Keras • Jupyter • Google Colab
• Git & GitHub

Program Outcome: Learners build practical skills across the complete AI/ML workflow—from data preparation and model development to evaluation, Generative AI, deployment, and presentation of a real-world project.